

Hierarchical classification structure based on SVM for volcano seismic events.

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Abstract— Volcano monitoring is very important in many countries. Chile is located in the Pacific fire belt, so the country has hundreds of volcanoes along its territory that may affect the economic and social activity. The main variable to monitor is volcano seismicity, as the recognition of seismic patterns originated inside the volcano gives important information about its activity. The great number of volcanos and the huge amount of seismic records makes necessary to automate some steps of this process. This paper presents a hierarchical structure based on Support Vector Machines (SVM) to classify four classes of seismic events recorded by the seismic stations. The implementation of the structure was performed using the database of the Llaima volcano and the classifiers were tested in another volcano, the Nevados de Chillán Volcanic Complex. The results show a good performance with a mean accuracy of 89.7% for the Llaima volcano and 84.9% for the Chillán volcano, with no retraining process for this new seismic dataset.

Keywords—volcano monitoring, seismic pattern recognition, support vector machines.

I. INTRODUCTION

The volcanic activity is mainly monitored by seismic sensors that record the movements caused by the fluids dynamics inside the ducts structure of a volcano. Many different kind of processes may lead to seismic events, generating classes of events that can be distinguished by characteristic patterns in terms of amplitude, frequency, duration etc. Recognizing these classes is important for monitoring as they are related to the volcano internal processes [1]. However, this is a difficult task because the geometry of the source, the propagation paths between the seismic source and the type of seismic stations and noise may affect the shape of the recorded seisms. This is why events classification is generally performed by human analysts who manually extract and label the events.

The Observatorio Vulcanológico de los Andes Sur (OVDAS) is the Chilean agency dependent on the National Geology and Mining Service responsible to monitor and report the activity of 43 of the most active volcanos. Because of the great amount of records OVDAS needs to automatize the seismic events classification.

A literature review shows that the volcano seismic events classification is not trivial, this is why many pattern recognition techniques are being applied [2]. The traditional structure is based on a feature extraction step and a classifying step. For the first many features have been selected in different works, extracted directly from the raw signals [3] or from their spectral representation [4]. In another hand, many techniques have been applied to implement the classifiers [5], including recently deep learning [6]. Although the results of previous works are encouraging, the differences between the volcanoes make it difficult to implement a universal seismic pattern recognition, so the research for better and flexible classifiers is ongoing.

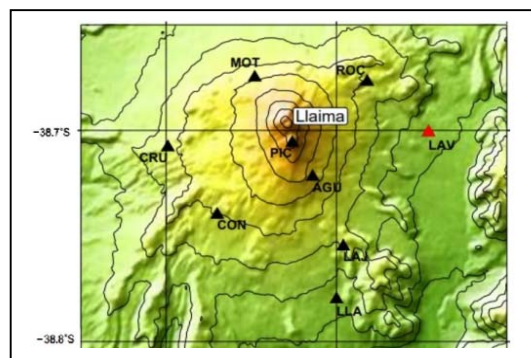


Fig. 1. Llaima volcano and its seismic stations (Δ).

In this work, a hierarchical classifying structure is proposed as an alternative to the classifying structures used in previous works [7-9] for the Llaima volcano. The motivation was to divide the multiple classes discrimination problem into steps in which discrimination is easier, contrasting with the "one versus all" structures previously defined. In this way, a tree-shaped classification structure was built, which allows classifying groups of events and finally classes in a hierarchical way. The structure was defined by a study performed on the features extracted from the seismic events. The features discrimination ability was used to decide the levels and groups considered for the hierarchical structure.

The paper presents the preliminary results of the hierarchical structure and compare the results with previous works. The performance of the classifying structure was also tested in another volcano to evaluate its robustness.

II. MATERIALS

A. Datasets

Two seismic events datasets were considered: one from the Llaima volcano (see Fig.1), located in the Araucanía region of Chile (S 38°41' - W 71°44'), one of the most hazardous volcanoes of south America, which was active until 2009 and therefore was studied heavily [7-9] and the Nevados de Chillan Volcanic Complex, located in the Ñuble region of Chile (S 36°52'-W 71°22'), which is an interesting case study for its recent eruptive activity.

The proposed method discriminates four seismic event classes from the records (see examples of the classes and their spectrograms in Fig. 2). The volcanic events classes are **long period** (LP) caused by the sudden transit of magmatic and hydrothermal fluids inside the volcanic conduits, while the **volcano-tectonic** (VT) are events associated to rock fracturing in the surrounding crust. In addition the stations record other events, not related to the volcanic activity such as **tectonic** (TC) events, which are produced by the dynamic of the geological faults and **other** (OT) events related to external phenomena like ice breaking due to the melting glaciers or events due to climate conditions, like wind, storms, etc. or also undefined noise. In addition, the **false positive** (FP) class is formed by records which are not seismic events and that have been erroneously selected by the segmentation process performed by an automatic detector of events. In this work, FP events are included in the OT class. Table 1 shows the number of events per class.

To design the classifiers, only events from the LAV seismic station (see Fig. 1) that had a SNR superior to 0.4 $\mu\text{m/s}$ and a duration greater than 5 seconds were stored in the Llaima dataset. All the events were sampled at 100Hz. In all cases, a 6-folds cross-validation was applied to design the classifiers.

To test the proposed classifying structure the events recorded from the Fresco (FRE) station of the Nevados de Chillán Volcanic Complex were considered. Table I shows the events

per class for both datasets. Notice that the TC class is not present in the Chillán dataset.

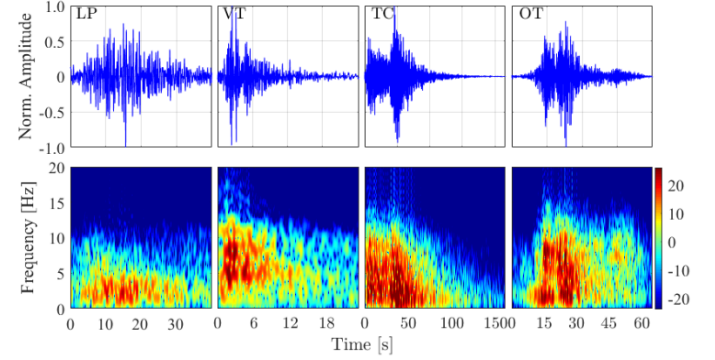


Fig. 2. Examples of the raw signal and the spectrogram of the 4 seismic event classes.

TABLE I. NUMBER OF EVENTS PER CLASS FOR THE DATASETS

	LP	VT	TC	OT	TOTAL
Llaima	210	51	462	267	990
Chillán	1824	126	-	22	1972

B. Feature extraction

The amplitude of the signals ($\mu\text{m/s}$) were normalized and a 10th order Butterworth band pass filter between 1 to 10 Hz was applied. The classifying hierarchical structure was designed from the analysis of a pool of 52 features extracted from each event. The features are related to signal amplitude, energy, frequency and statistical measures, which are described below.

- Ax: Maximum value of the amplitude obtained from the records in the time domain.
- An: Average value of the amplitude obtained from the time domain.
- Duration (D): Corresponds to the duration of each event in seconds from P wave arrive.
- E_n: Percentage of energy in the nth of the 7 bands, obtained from the Wavelet transform.
- Dominant frequency (DF): Frequency of the maximum peak of magnitude of the spectrum.
- Pitch (P): Related to the lowest frequency at which an object vibrates. In speech processing it is the perception of the acoustic manifestation of intonation [10].
- LPC_n: Linear Predictive Coding Coefficients [11] represents the signal spectral envelope in a compressed form. For each signal, five LPC were calculated.
- F₅: Average energy of the five highest frequency peaks obtained from magnitude spectrum.
- Statistical measures of E_n: Each event is segmented using a rectangular window of 250 samples. For each window, E_n is calculated obtaining the percentage energy in the nth window in time over the total energy. From this vector the variance (VarE_n), skewness (SkwE_n) and kurtosis (KrtE_n) are computed.

- Statistical measures of F_5 : The spectral magnitude of each event is segmented using a rectangular window of 150 samples. F_5 is calculated for each window. From this vector the average (AvF_5), variance ($VarF_5$), skewness ($SkwF_5$) and kurtosis ($KrtF_5$) are computed.
- Statistical measures of the Pitch: The spectral magnitude of each event is segmented using a rectangular window of 150 samples. The Pitch is calculated for each window. From this vector the average (AvP), variance ($VarP$), skewness ($SkwP$) and kurtosis ($KrtP$) are computed.
- Statistical measures of DF: The spectral magnitude of each event is segmented using a rectangular window of 64 samples. DF is calculated for each window. From this vector the average ($AvDF$), variance ($VarDF$), skewness ($SkwDF$) and kurtosis ($KrtDF$) are computed.

Some features were extracted from a spectro-temporal analysis described in a previous work [8] where two temporal curves were defined to characterize the spectrograms of the events. The curves are the spectral evolution (SE) and the energy accumulation ($\hat{\rho}$). Two features were extracted from these curves:

- Percentage Rise time (RT_{i-f}): The energy accumulation $\hat{\rho}$ curve [8] is calculated. Then, the RT_{i-f} is calculated as the rise time from the initial point (i) to the final point (f) of the $\hat{\rho}$ curve, over the event duration.
- SEA: Corresponds to the percentage of the spectral evolution curve (SE) [8] area. It is calculated as $SEA = SE_{area}/T_{area}$, where SE_{area} is the area under the curve, and T_{area} is total area given by a rectangle of width equals to the curve length and height equals to its maximum value.

III. METHOD

A. Classifying hierarchical structure

The classification structure is based on Support Vector Machines (SVM) [12] with a RBF kernel. Since SVM only discriminate two classes, in our previous works the classifiers were organized in a “one versus all” structure, all receiving, in parallel, the characteristics of an event and retrieving the output of all the classifiers. It was therefore necessary to have a process of reliability evaluation to select the label of the most reliable classifier.

Unlike previous works, here the classifiers perform a hierarchical discrimination in the form of a tree, as shown in Fig. 3. In this tree, the classification is carried out in different levels, where, at each level, the classifiers must decide between two groups of classes or two classes. To train the classifiers of each level, the best combination of all the features was evaluated, using the method described next. Notice in Fig. 3 that the structure defined subgroups of the TC and OT classes (TC_1 , TC_2 , OT_1 and OT_2). Table II presents the features selected for each level. The hierarchical structure was defined by a statistical analysis that evaluated the discrimination ability of all the features. To do so, individual feature graphs and

histograms were analyzed. Each feature was also divided in two clusters, using the K-means clustering technique. The clusters were compared with the classes and the discrimination ability was computed for each class. The different levels of the hierarchical structure were established according to the best clusters obtained from the individual analysis of all the features.

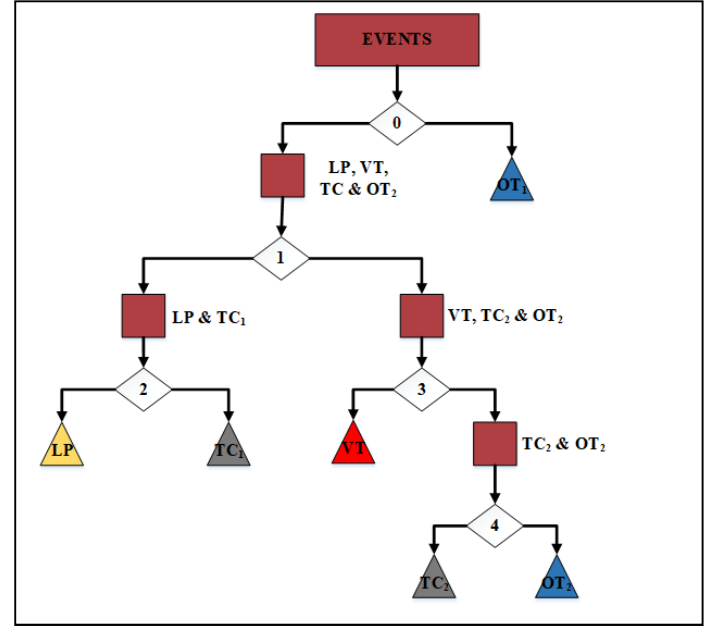


Fig. 3. Hierarchical structure of the SVM classifiers.

In order to establish each level, the following criteria were considered:

- Evaluate how well each feature separates the classes or group of classes.
- Identify the clusters that occurred more often when analyzing the individual features.
- Preferably consider the clusters that better separate the volcanic classes (LP and VT).

This procedure was performed for each level, excluding the events separated by the previous level, using only the remaining events to define a new division. Therefore, the development of the classification structure was sequential.

For the training of each division, a maximum of 10 features were selected that presented the best discrimination ability. From this subset, the best combination of up to 4 characteristics was sought, performing the procedure described in section B to train the classifiers.

TABLE II. SELECTED FEATURES FOR EACH CLASSIFYING LEVEL

Level	Feature vector
0	[$VarE_4$, $SkwP$, F_5]
1	[E_1 , E_2 , AvP , $AvDF$]
2	[D , $VarE_3$, F_5 , SEA]
3	[D , RT_{5-95} , RT_{2-30}]
4	[D , E_7 , $VarE_5$, LPC_3]

B. Training process of the SVM classifiers

To train the SVM classifiers it is necessary to tune two hyperparameters: c is the constant that defines the model regularization, that is, the trade-off between its complexity and its empirical error. σ is the constant that defines the RBF Gaussian. To do so, a grid search method was performed where c is expressed as 2^a and σ as 2^b . Therefore the scan of c was performed by varying the factor a from -5 to 15, and for σ the factor b varied from -15 to 10, both variations with a step of 2. Once the training of all the hyperparameters combinations was carried out, the best one was selected from the result obtained with the test database.

C. Performance Indices

The following statistical measures were defined based on the results of the confusion matrix of two classes, where one is considered the positive class and the other the negative class [13]. When the number of classes is superior, the indices are calculated for each class considering it the positive and all the remaining classes form the negative class.

True positives ratio (TPR) or sensitivity, indicates the percentage of positive class elements that were correctly classified. True negatives ratio (TNR) or specificity, indicates the percentage of elements of the negative class that were correctly classified. False discovery ratio (FDR) is the percentage of the elements incorrectly classified as positive. Positive predictive value (PPV) indicates the reliability of the positive class classification. Accuracy is the percentage of events correctly classified while Error is the percentage of elements misclassified. Each index is defined in the equations from 1 to 6, where TP and TN are the number of true positive and true negative events, FP and FN the number of false positive and false negative events and n is the total number of events.

$$TPR = \frac{TP}{TP + FN} 100 \quad (1)$$

$$TNR = \frac{TN}{TN + FP} 100 \quad (2)$$

$$FDR = \frac{FP}{TP + FP} 100 \quad (3)$$

$$PPV = \frac{TP}{TP + FP} 100 \quad (4)$$

$$Accuracy = \frac{TP + TN}{n} 100 \quad (5)$$

$$Error = \frac{FP + FN}{n} 100 \quad (6)$$

IV. RESULTS

In this section the performance of the hierarchical classifying structure will be presented. Tables III and IV show the good performance of the classifiers for the Llama dataset. In particular the LP, VT and TC classes are well discriminated (mean Accuracy 89.7%) and the sensitivity and specificity of the classifiers is high (mean of TPR 79.97% and mean TNR 93.15%).

Table V compares these results with other works on the Llama volcano. In [7] for the first time we proposed a “one Vs all” structure applied to the Llama volcano dataset with a simple feature extraction. The TC class was not considered in this work. In [8] the SVM “one Vs all” structure was applied using more complex features, most of which are used in the present work. Notice that in this case the OT class is not present. Finally, the results of the hierarchical structure are compared with a recent work where a deep CNN was used to perform the discrimination [9].

TABLE III. CONFUSION MATRIX FOR THE LLAMA DATASET

Hierarchical classifiers						
Expert		LP	VT	TC	OT	Total
	LP	188	2	10	10	210
	VT	5	43	2	1	51
	TC	8	5	391	58	462
	OT	28	56	19	164	267
	Total	229	106	422	233	990

TABLE IV. PERFORMANCE INDICES FOR THE LLAMA DATASET

Index	LP	VT	TC	OT	Mean
TPR	89,52	84,31	84,63	61,42	79,97
TNR	94,74	93,29	94,13	90,46	93,15
FDR	17,90	59,43	7,35	29,61	28,57
PPV	82,10	40,57	92,65	70,39	71,43
Accuracy	93,64	92,83	89,70	82,63	89,70
Error	6,36	7,17	10,30	17,37	10,30

Table V shows the sensitivity (TPR), the error and the number of events (n) per class for each work. It can be observed that there is no significant improvement in the classification performance of the hierarchical structure when compared with previous works. However, a special analysis has to be performed on the OT class that contains a very difficult to classify set of seismic events. The OT class has events whose origin may be caused by a great variety of sources, so the characteristics of these events are very heterogeneous, what produces great intra-class variability and in some cases a great similarity to events of other classes. As shown by the TPR for the [7] and [9] references, the sensitivity is low for this class. The hierarchical structure was able to maintain a performance superior to 80% with a sensitivity of 61.4%. This result is interesting as the automatic segmentation of events generates a high number of FP that were considered in the OT class.

Tables VI and VII show the results of the hierarchical structure when applied to classify a new dataset: the Nevados del Chillán volcanic complex. This is an interesting result as the model was not trained using the Chillán dataset and this volcano has a different structure and composition when compared to the Llama volcano. The results show a good accuracy (84.9%) but a low sensitivity for the OT and VT classes. However, the

resulting model was robust enough to reach a good performance with an unknown volcano. This is an encouraging result to evaluate the generalization ability of the proposed classifying structures, what is absolutely necessary to expand the results of this research to other Chilean volcanoes.

TABLE V. PERFORMANCE INDICES FOR THE PREVIOUS SVM “ONE VS ALL” [7- 8] AND CNN [9] STRUCTURES AND PRESENT WORK

		LP	VT	TC	OT	MEAN
[7]	TPR	86,34	85,71	-	68,94	80,33
	Error	11,28	2,82	-	18,23	10,78
	n	499	148	-	475	
[8]	TPR	96,79	93,14	98,09	-	95,62
	Error	1,73	1,52	1,12	-	1,46
	N	746	304	1411	-	
[9]	TPR	97,71	95,39	93,75	37,43	81,07
	Error	2,34	2,13	3,67	4,15	3,07
	N	6835	2471	1863	723	
Present	TPR	89,52	84,31	84,63	61,42	79,97
	Error	6,36	7,17	10,30	17,37	10,30
	N	210	51	462	267	

TABLE VI. CONFUSION MATRIX FOR THE CHILLÁN DATASET

Hierarchical classifiers					
Expert		LP	VT	OT	Total
	LP	1444	205	175	1824
	VT	39	72	15	126
	OT	7	5	10	22
	Total	1490	282	200	1972

TABLE VII. PERFORMANCE INDICES FOR THE CHILLÁN DATASET

Index	LP	VT	OT	MEAN
TPR	79,17	57,14	45,45	60,59
TNR	68,92	88,62	90,26	82,60
FDR	3,09	74,47	95,00	57,52
PPV	96,91	25,53	5,00	42,48
Accuracy	78,40	86,61	89,76	84,92
Error	21,60	13,39	10,24	15,08

V. CONCLUSIONS

This paper presents a new proposal for the classifying structure used to label four classes of events recorded by the seismic stations that monitor the Llaima volcano. The purpose of the hierarchical architecture was to simplify the effort of the classifiers. The relations of the features and the classes were analyzed to define the specialized classifiers used in each level of the hierarchical structure.

This is a relevant purpose as the noise and FP events decrease the performance of the “one Vs all” classifiers. The different levels of the hierarchical structure were consistent with the difficulty to discriminate each class. The TC and the OT

classes required many steps to be classified, as shown in Fig.3. This is due to the fact that these classes have events with very heterogeneous shapes, because their origin are different. The TC characteristics depend on the distance between the epicenter and the recording station. Local or far TC can be confused with VT or OT classes, as shown in Table III. For the OT class, the variability is even higher, and the events may be confused with all the classes as shown in Table III.

The hierarchical structure was intended to improve the classification performance of the classifiers when compared with previous works. However, even if the performance is good, the proposed structure did not significantly improve the previous results. Despite of this, the performance was good for the OT compared to previous works. This is why it still represents an interesting alternative.

The test in another volcano was considered good as the accuracy was superior to 80% with no retraining process. It is important to notice that the performance of each class had a similar behavior when compared with the Llaima performance classes: the evaluation was considered consistent in both databases.

Finally it is important to underline that the preliminary results presented in this work are encouraging to go on searching new methods to define the hierarchical levels and to get a structure able to tackle the discrimination problem separating the whole problem into smaller and easier to discriminate sub-problems, applying the “divide to conquer” strategy.

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